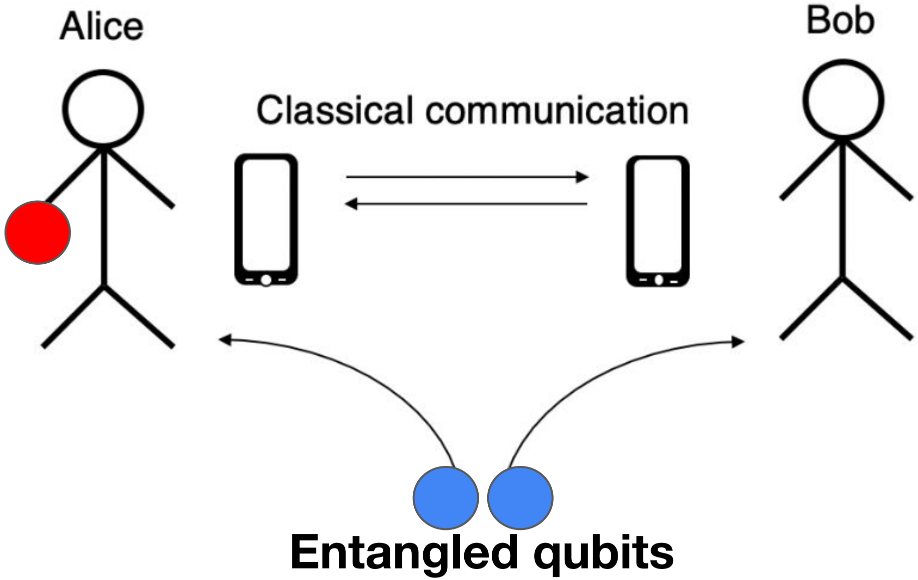
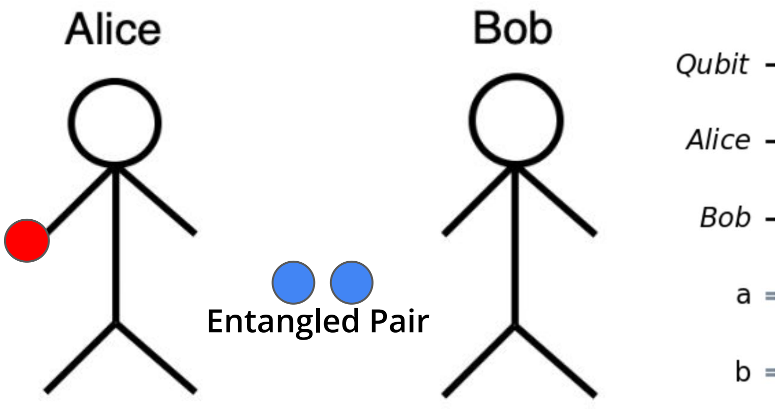
Quantum Teleportation

Teleportation – Quantum Information Transfer



- **Alice transfers a qubit state to Bob using entanglement**
- **Neither Alice nor Bob knows the state of this qubit**
- **This protocol requires Alice to communicate classical information to Bob (0s and 1s)**

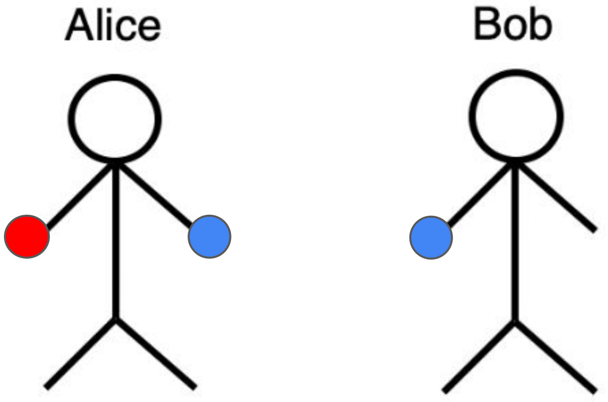
Initial State Preparation

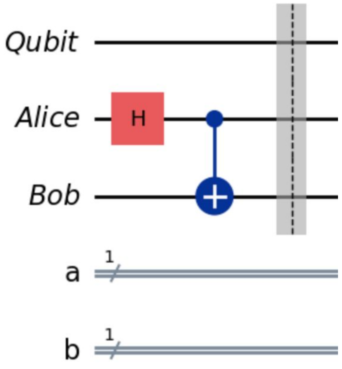



```
qubit = QuantumRegister(1, "Qubit")
ebit0 = QuantumRegister(1, "Alice")
ebit1 = QuantumRegister(1, "Bob")
a = ClassicalRegister(1, "a")
b = ClassicalRegister(1, "b")
protocol = QuantumCircuit(qubit, ebit0, ebit1, a, b)
```

[Reference: IBM Quantum Learning]

Shared Entanglement

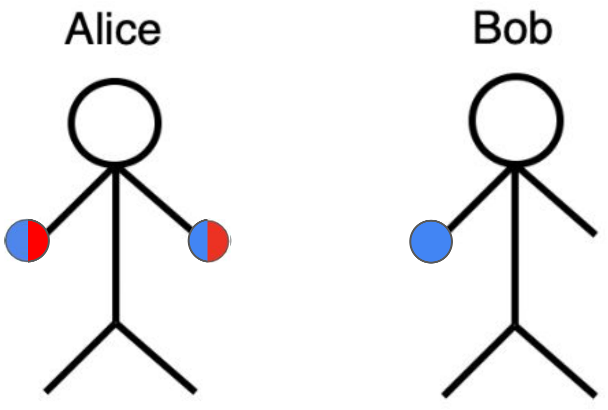


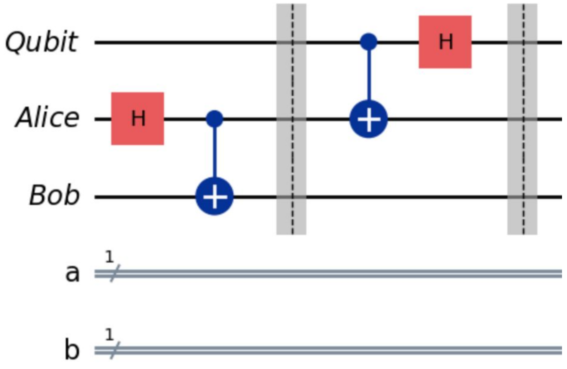



```python
# Prepare entangled qubits used for teleportation
protocol.h(ebit0)
protocol.cx(ebit0, ebit1)
protocol.barrier()
```

[Reference: IBM Quantum Learning]

Alice Entangles Her Qubit

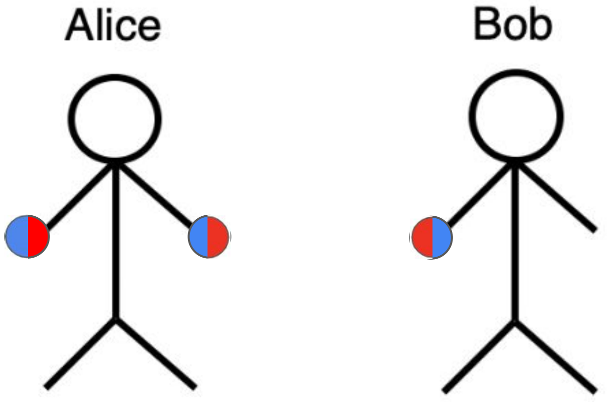


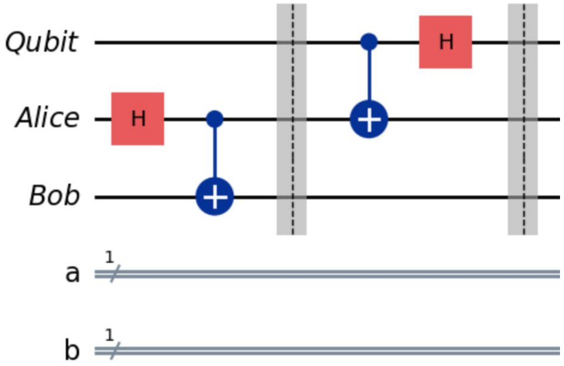



```
# Alice's operations
protocol.cx(qubit, ebit0)
protocol.h(qubit)
protocol.barrier()
```

[Reference: IBM Quantum Learning]

Entanglement ⇒ Information Transferred to Bob

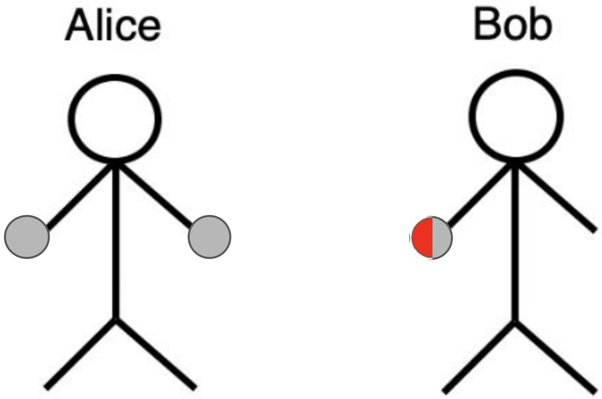


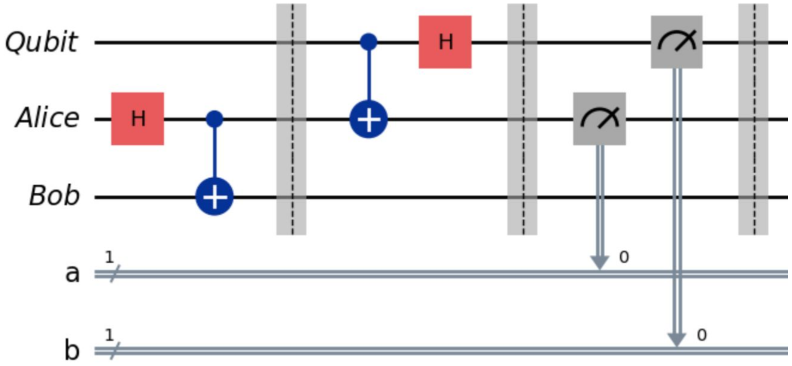



```
# Alice's operations
protocol.cx(qubit, ebit0)
protocol.h(qubit)
protocol.barrier()
```

[Reference: IBM Quantum Learning]

Alice Measures Her Qubits

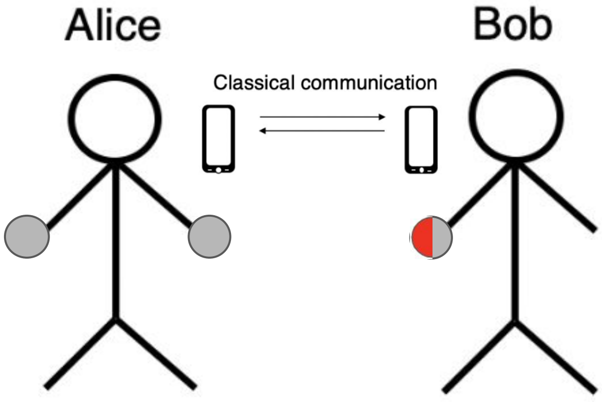


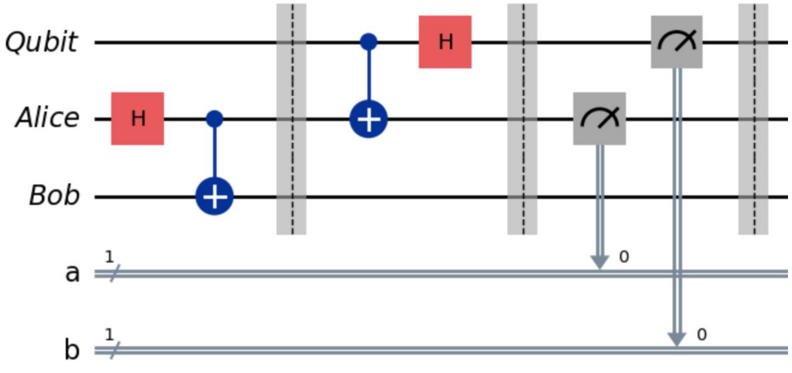



```python
# Alice measures and sends classical bits to Bob
protocol.measure(ebit0, a)
protocol.measure(qubit, b)
protocol.barrier()
```

[Reference: IBM Quantum Learning]

Alice Communicates Her CLASSICAL Info to Bob

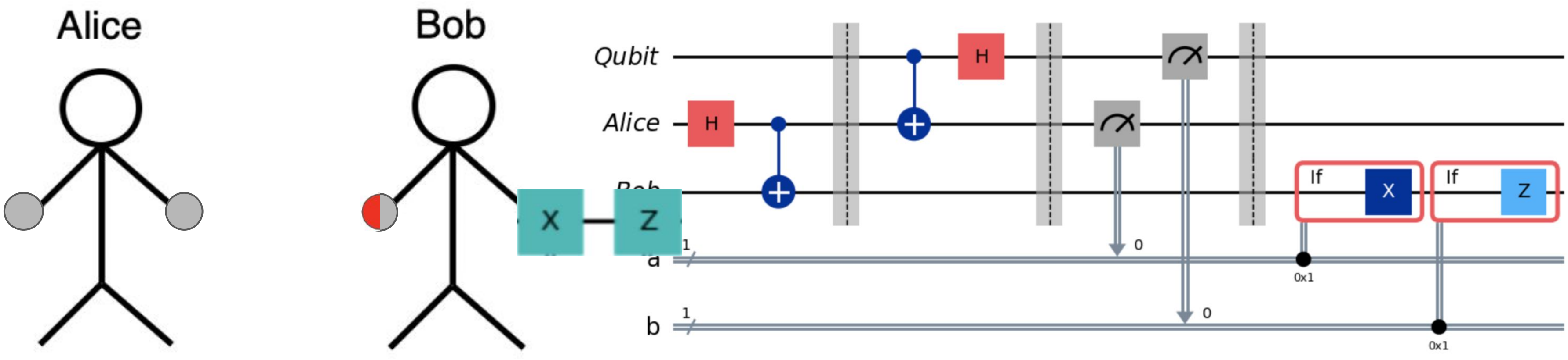





```
# Alice measures and sends classical bits to Bob
protocol.measure(ebit0, a)
protocol.measure(qubit, b)
protocol.barrier()
```

[Reference: IBM Quantum Learning]

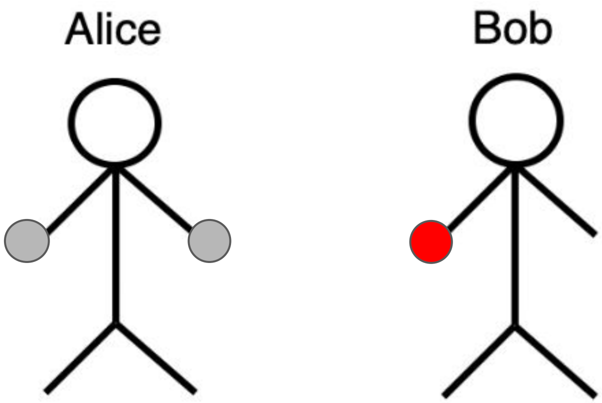
Bob Alters His Qubits Based on Alice’s Measurements

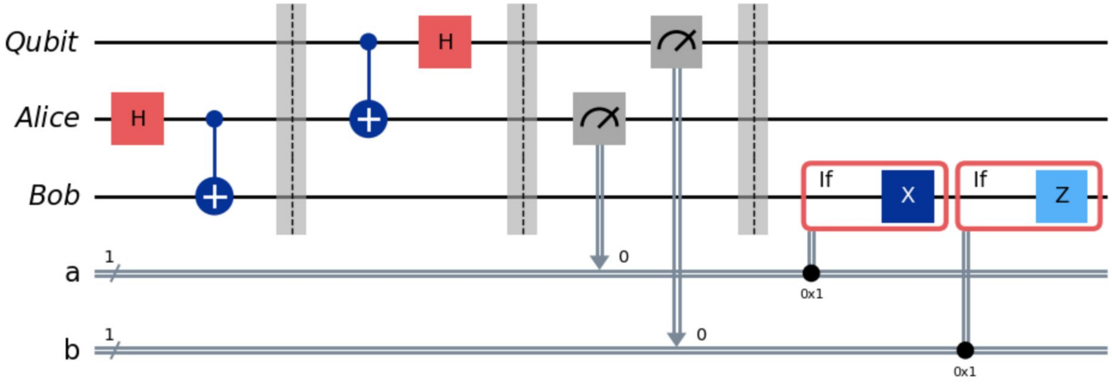



```python
# Bob uses classical info to conditionally apply gates
with protocol.if_test((a, 1)):
   protocol.x(ebit1)
with protocol.if_test((b, 1)):
   protocol.z(ebit1)
```

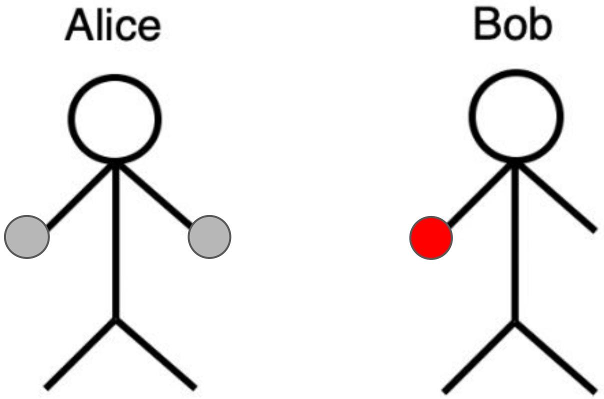
[Reference: IBM Quantum Learning]

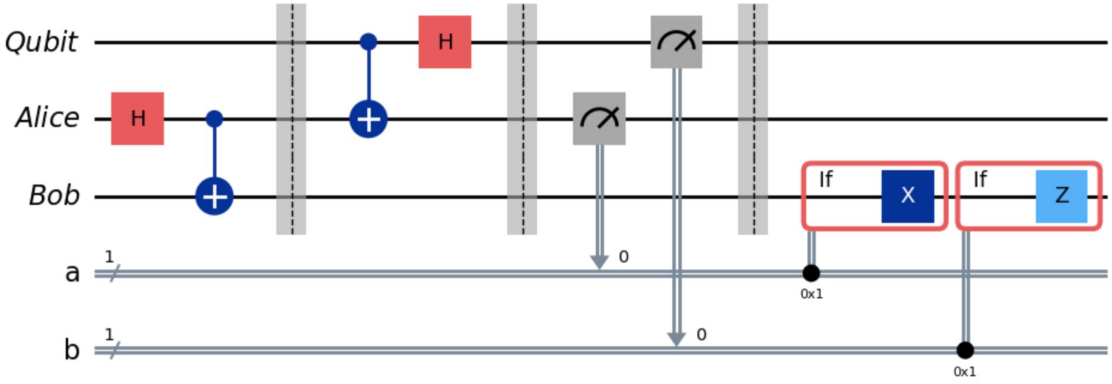
Bob Recovers Alice's Original Qubit State





Takeaways From Teleportation



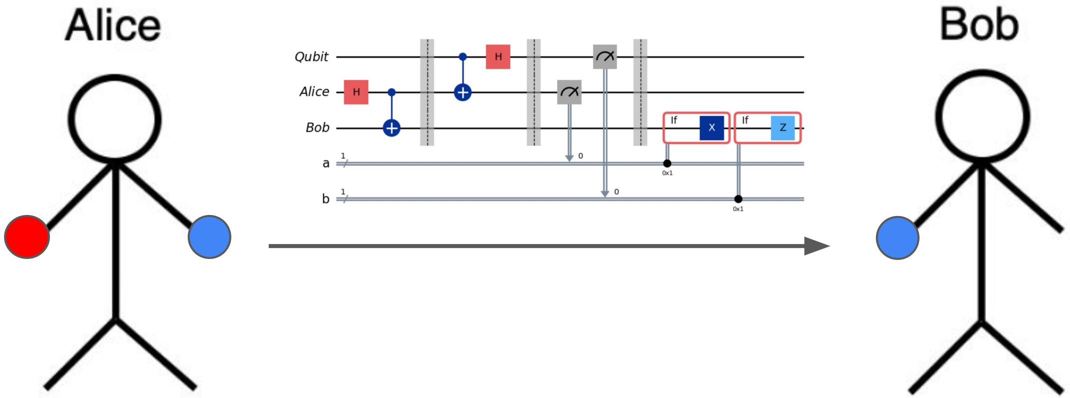


- Neither Alice nor Bob needed to know the initial qubit state
  ⇒ Requires no knowledge of the information being transferred

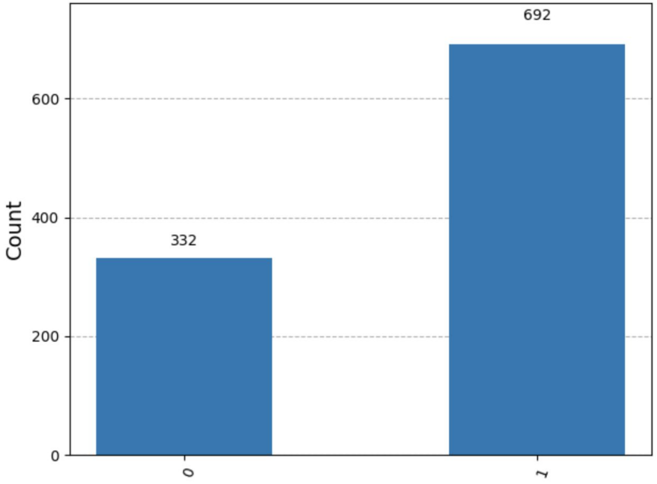
- Only classical information is transferred between Alice and Bob
  ⇒ An eavesdropper would gain no useful quantum information

- Alice's original qubit state was transferred, NOT cloned, onto Bob's qubit

Teleportation in Practice



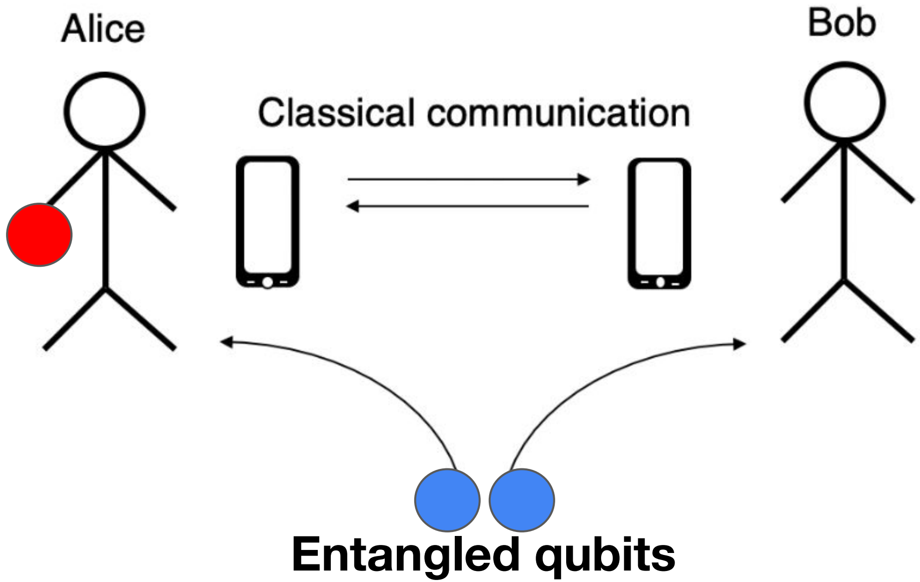
$$|\psi\rangle_{\text{Alice}} = \frac{1}{\sqrt{3}}|0\rangle + \sqrt{\frac{2}{3}}|1\rangle$$



IBM's Simulator

BACKUP

Teleportation – Quantum Information Transfer



Initial States

- Alice has one qubit
- Alice and Bob share a pair of entangled qubits

Information Entanglement

- Alice entangles her qubit with one qubit in the pre-entangled pair
- The entanglement transfers onto the other qubit in the pre-entangled pair, held by Bob

The Final State

- Alice communicates measurements of her qubits to Bob
- Bob uses that information to recover Alice's original qubit state